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A Model for Measurement Error

Upon the whole of which it appears, that the taking of the Mean of a number of observations, greatly diminishes the chance for all the smaller errors, and cuts off almost all possibility of any great ones: which last consideration, alone, seems sufficient to recommend the use of the method, not only to astronomers, but to all others concerned in making experiments of any kind (to which the above reasoning is equally applicable). And the more observations or experiments there are made, the less will the conclusions be liable to error, provided they admit of being repeated under the same circumstances.

—THOMAS SIMPSON (ENGLISH MATHEMATICIAN, 1710–1761)

1. ESTIMATING THE ACCURACY OF AN AVERAGE

In this part of the book, the frequency theory of chance will be used to study measurement error and genetics. Historically, the frequency theory was developed to handle problems of a very special kind—figuring the odds in games of chance. Some effort is needed to apply the theory to situations outside the gambling context. In each case, it is necessary to show that the situation being studied resembles a process—like drawing at random from a box—to which the theory applies. These box models are sometimes called *chance models* or *stochastic models*. The first example will be a chance model for measurement error.

To review briefly (chapter 6), any measurement is subject to chance error, and if repeated would come out a bit differently. To get at the size of the chance error, the best thing to do is to repeat the measurement several times. The spread

in the measurements, as shown by the SD, estimates the likely size of the chance error in a single measurement. Chapter 6 stopped there. This chapter continues the discussion: the focus is on the average of the measurements in the series rather than a single measurement. The problem is to estimate the likely size of the chance error in the average. If the measurements are like draws from a box, the methods of parts V and VI can be used.

Table 1 on p. 99 shows 100 measurements on NB 10. These all fell short of 10 grams, by different amounts. The table gives the amounts, in micrograms. (A microgram is one millionth of a gram, roughly the weight of a speck of dust.) The SD of the 100 numbers in the table is about 6 micrograms: a single measurement is only accurate up to 6 micrograms or so. The best guess for the weight of NB 10 is the average of all 100 measurements, which is 404.6 micrograms short of 10 grams. Since each measurement is thrown off by error, the average cannot be exactly right either. But the average is going to be more accurate than any single measurement, so it is going to be off by less than 6 micrograms.

What is the right give-or-take number to put on the average?

$$\text{average} \pm _____.$$

The answer is given by the SE for the average, which can be calculated just as in chapter 23. (The calculation rides on a box model, to be discussed in sections 2 and 3 below.) The SE for the sum of 100 measurements can be estimated as

$$\sqrt{100} \times 6 \text{ micrograms} = 60 \text{ micrograms}.$$

Then the SE for the average of the 100 measurements is

$$\frac{60 \text{ micrograms}}{100} = 0.6 \text{ micrograms}.$$

This completes the calculation. The average of all the numbers in the table is 404.6 micrograms. The likely size of the chance error in the average is estimated to be 0.6 micrograms. So NB 10 really weighs about 404.6 micrograms below 10 grams, plus or minus 0.6 micrograms or so.

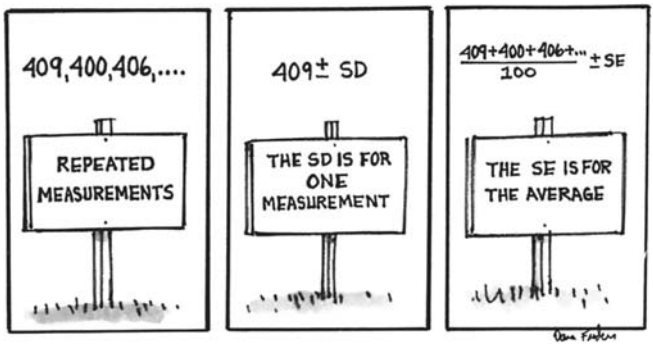
Two numbers come up in the calculation: 6 micrograms and 0.6 micrograms. The first is the SD of the 100 measurements, the second is the SE for the average. What is the difference between the two?

- The SD says that a single measurement is accurate up to 6 micrograms or so.
- The SE says that the average of all 100 measurements is accurate up to 0.6 micrograms or so.

Example 1. One hundred measurements are made on a certain weight. The average of these measurements is 715 micrograms above one kilogram, and the SD is 80 micrograms.

(a) Is a single measurement likely to be off the exact weight by around 8 micrograms, or 80 micrograms?

(b) Is the average of all 100 measurements likely to be off the exact weight by around 8 micrograms, or 80 micrograms?



Solution. A single measurement is off by a chance error similar in size to the SD of the measurements. This is 80 micrograms. The answer to (a) is 80 micrograms. For (b), the SE for the sum of the measurements is estimated as $\sqrt{100} \times 80 = 800$ micrograms. So the SE for the average is $800/100 = 8$ micrograms. That is the answer to (b).

In example 1, the give-or-take number for the average of the measurements is 8 micrograms. To make this more precise, statisticians use confidence intervals, just as in sampling. A 95%-confidence interval for the exact weight can be obtained by going 2 SEs in either direction from the average. The average is 715 micrograms above one kilogram, and 2 SEs is $2 \times 8 = 16$ micrograms. So the exact weight is somewhere between 699 and 731 micrograms above one kilogram, with confidence about 95%. The arithmetic:

$$715 - 16 = 699, \quad 715 + 16 = 731.$$

Again, the word "confidence" is there as a reminder that the chances are in the measurement process and not in the thing being measured: the exact weight is not subject to chance variability. (For a similar discussion in the sampling context, see section 3 of chapter 21.)

> The chances are in the measuring procedure, not the thing being measured.

The normal curve should be used to get confidence intervals only when there is a fairly large number of measurements. With fewer than 25 measurements, most statisticians would use a slightly different procedure, based on what is called the t-distribution (section 6 of chapter 26).

Historical note. There is a connection between the theory of measurement error and neon signs. In 1890, the atmosphere was believed to consist of nitrogen (about 80%), oxygen (a little under 20%), carbon dioxide, water vapor—and nothing else. Chemists were able to remove the oxygen, carbon dioxide, and water vapor. The residual gas should have been pure nitrogen.

Lord Rayleigh undertook to compare the weight of the residual gas with the weight of an equal volume of chemically pure nitrogen. One measurement on the weight of the residual gas gave 2.31001 grams. And one measurement of the pure nitrogen gave a bit less, 2.29849 grams. However, the difference of 0.01152 grams was rather small, and in fact was comparable to the chance errors made by the weighing procedure.

Could the difference have resulted from chance error? If not, the residual gas had to contain something heavier than nitrogen. What Rayleigh did was to replicate the experiment, until he had enough measurements to prove that the residual gas from the atmosphere was heavier than pure nitrogen.

He went on to isolate the rare gas called *argon*, which is heavier than pure nitrogen and present in the atmosphere in small quantities. Other researchers later discovered the similar gases neon, krypton, and xenon, all occurring naturally (in trace amounts) in the atmosphere. These gases are what make "neon" signs glow in different colors.[1]

Exercise Set A

1. The total of the 100 measurements on NB 10 was 40,459 micrograms. What is the likely size of the chance error in this total?

2. Some scales use electrical *load cells*. The weight is distributed over a number of cells. Each cell converts the weight it carries to an electrical current, which is fed to a central scanner. This scanner adds up all the currents, and computes the corresponding total weight, which it prints out. This process is repeated several dozen times a second. As a result, a loaded boxcar (weighing about 100,000 pounds) can be weighed as it crosses a special track, with chance errors of only several hundred pounds in size.[2]

   Suppose 25 readings on the weight of a boxcar show an average of 82,670 pounds, and the SD is 500 pounds. The weight of the boxcar is estimated as _____; this estimate is likely to be off by _____ or so.

3. (Hypothetical.) The British Imperial Yard is sent to Paris for calibration against The Meter. Its length is determined 100 times. This sequence of measurements averages out to 91.4402 cm, and the SD is 800 microns. (A *micron* is the millionth part of a meter.)

   (a) Is a single reading off by around 80 microns, or 800 microns?
   (b) Is the average of all 100 readings off by around 80 microns, or 800 microns?
   (c) Find a 95%-confidence interval for the exact length of the Imperial Yard.

4. The 95%-confidence interval for the exact weight of NB 10 is the range from 403.4 to 405.8 micrograms below 10 grams. Say whether each of the following statements is true or false, and explain why.

   (a) About 95% of the measurements are in this range.
   (b) There is about a 95% chance that the next measurement will be in this range.
   (c) About 95% of the time that the Bureau takes 100 measurements and goes 2 SEs either way from the average, they succeed in covering the exact weight.
   (d) If the Bureau took another 100 measurements on NB 10, there is about a 95% chance that the new average would fall in the interval from 403.4 to 405.8 micrograms below 10 grams.

5. Would taking the average of 25 measurements divide the likely size of the chance error by a factor of 5, 10, or 25?

The answers to these exercises are on p. A88.

2. CHANCE MODELS

Section 1 explained how to put a standard error on the average of repeated measurements. The arithmetic is easily carried out on any list of numbers, but the method is legitimate only when the variability in the data is like the variability in repeated draws from a box.

> If the data show a trend or pattern over time, a box model does not apply.

The reason: draws from a box do not show a trend or pattern over time. The following examples illustrate this idea.

Example 2. Table 1 gives the population of the U.S. from 1790 to 2000. Do these numbers look like draws at random from a box?

Table 1. Population of the U.S., 1790 to 2000.

1790	3,929,214
1800	5,308,483
1810	7,239,881
1820	9,638,453
1830	12,866,020
1840	17,069,453
1850	23,191,876
1860	31,443,321
1870	39,818,449
1880	50,189,209
1890	62,979,766
1900	76,212,168
1910	92,228,496
1920	106,021,537
1930	123,202,624
1940	132,164,569
1950	151,325,798
1960	179,323,175
1970	203,302,031
1980	226,542,199
1990	248,718,302
2000	281,422,602

Notes: Resident population. From 1950 onwards, includes Alaska and Hawaii. Revised figures for 1870–1940. Source: *Statistical Abstract*, 2006, Table 1.

Solution. No. The population of the U.S. has been going up steadily. Numbers drawn at random from a box don't do that: sometimes they go up and other times they go down.

Example 3. The 22 numbers in table 1 average out to 94.7 million, and the SD is 89.3 million. An investigator attaches a standard error to the average, by the following procedure:

$$\text{SE for the sum} \approx \sqrt{22} \times 89.3 \text{ million} \approx 419 \text{ million}$$
$$\text{SE for average} \approx 419/22 \approx 19.0 \text{ million.}$$

Is this sensible?

Solution. The average and SD make sense, as descriptive statistics. They summarize part of the information in table 1, although they miss quite a bit—for instance, the fact that the numbers increase steadily. The SE of 19 million, however, is silly. If the investigator wants to know the average of the 22 numbers in the table, that has been computed, and there is no need to worry about chance error. Of course, something else may be involved, like the average of a list showing the population of the U.S. in every year from 1790 to 2000. (Every tenth number on that list is shown in table 1; the numbers in between are known with less precision, because the Census is only taken every ten years.) The investigator would then be making an inference, using the average from table 1 to estimate that other average. And the estimate would be off by some amount. But the square root law cannot help much with the margin of error. The reason is that the numbers in table 1 are not like draws from a box.

The square root law only applies to draws from a box.

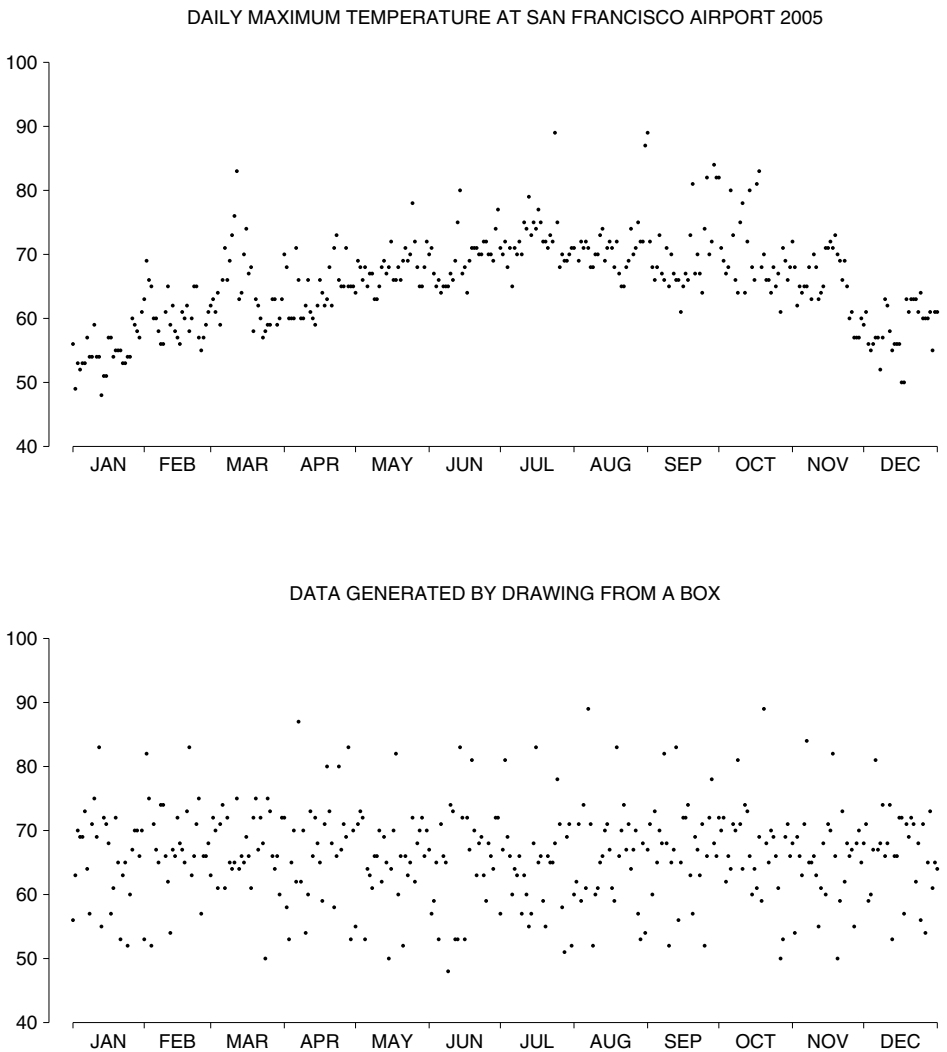
Example 4. A list is made, showing the daily maximum temperature at San Francisco airport. Are these data like draws from a box?

Solution. No, there is a definite seasonal pattern to these data—warmer in the summer, colder in the winter. There even are local patterns to the data. The temperature on one day tends to be like the temperature on the day before.

The temperature data are graphed in the top panel of figure 1. There is a dot above each day of the year for 2005, showing the maximum temperature on that day. The seasonal pattern is clear. On the whole, the dots are higher in the summer than in the winter. Also, there is an irregular wavy pattern within each season. The crest of a wave represents a stretch of warm days—a warm spell. The cold spells are in the troughs.

By comparison, the second panel in figure 1 is for a mythical airport where the climate is on average like San Francisco, but the daily maximum temperatures are like draws from a box. These data are random: they show no trend or pattern over the year. With this sort of climate, weather forecasting would be hopeless.

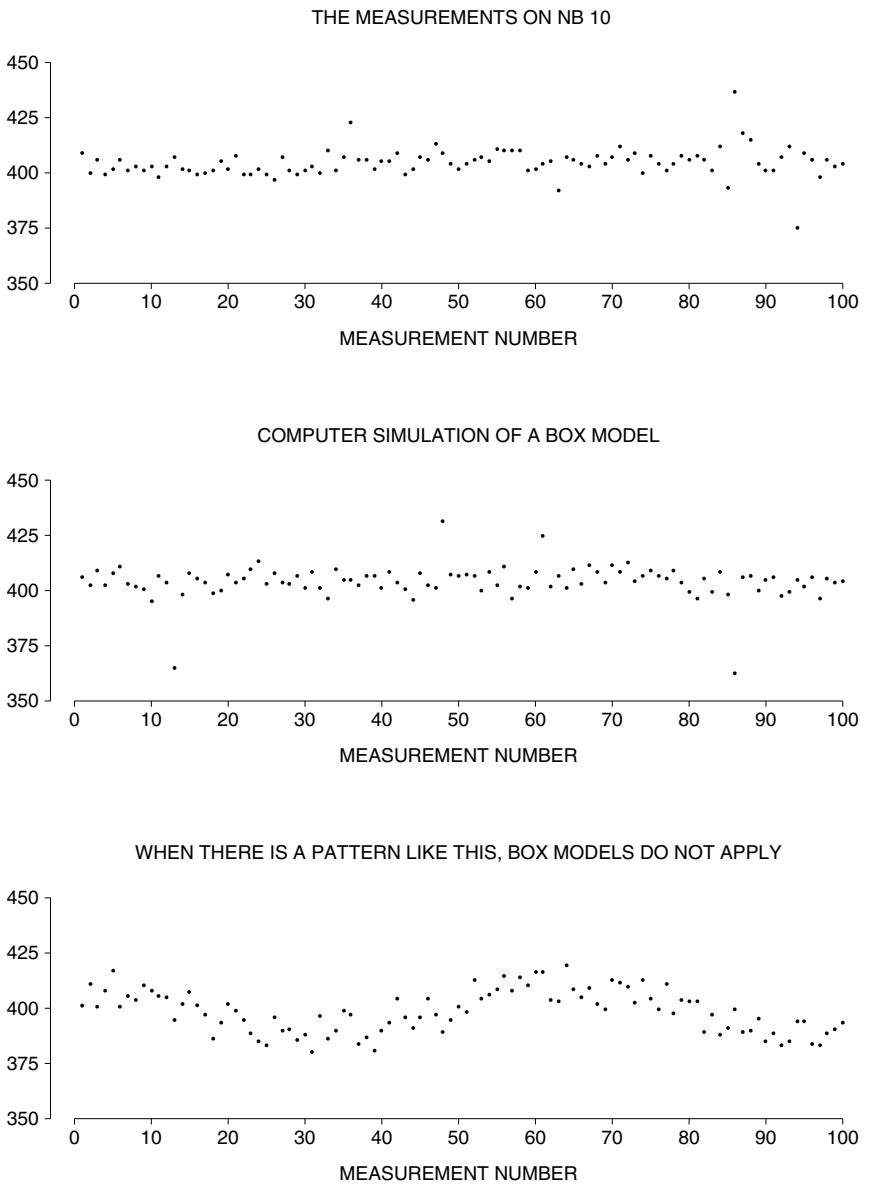
Figure 1. Temperature and box models. The first panel shows the daily maximum temperature at San Francisco airport in 2005.[3] There is a seasonal pattern to the data, warmer in summer than winter. Also, there are local patterns: warm spells and cold spells. A box model would not apply. The second panel shows what the temperatures would look like if they were generated by drawing from a box.



In section 1, we acted as if the measurements on NB 10 were like draws from a box. Was this sensible? The top panel in figure 2 (next page) is a graph of the data. There is one point for each measurement. The x-coordinate says which

measurement it was: first, or second, or third, and so on. The y-coordinate says how many micrograms below 10 grams the measurement was. The points do not show any trend or pattern over time; they look as random as draws from a box. In

Figure 2. The top panel graphs the repeated measurements on NB 10 (p. 99). The middle panel graphs hypothetical data, generated by computer simulation of a box model. These two panels are very similar, showing how well the box model represents the real data. The bottom panel graphs data showing a strong pattern: a box model would not apply.



fact, the second panel shows hypothetical data generated on the computer using a box model.[4] If you did not know which was which, it would be hard to tell the difference between these two panels. By comparison the third panel (also for computer-generated data) shows a strong pattern: a box model would not apply.

It is no accident that the data on NB 10 look like draws from a box. Investigators at the Bureau use pictures of the data, like the top panel in figure 2, to check their work. A trend or pattern is a signal that something is wrong and needs to be fixed. This idea is basic to precision measurement work—and to quality control in manufacturing, where the number of defective units is plotted against time.

Exercise Set B

1. A thumbtack is thrown in the air. It lands either point up or point down.

   Someone proposes the following box model: drawing with replacement from the box [U D], where U means "point up" and D means "point down." Someone else suggests the box [U D D]. How could you decide which box was better?

2. In San Francisco, it rains on about 17% of the days in an average year. Someone proposes the following chance model for the sequence of dry and rainy days: draw with replacement from a box containing one card marked "rainy" and five cards marked "dry." Is this a good model?

3. Someone goes through the phone book, and makes a list showing the last digit of each phone number. Can this be modeled by a sequence of draws (with replacement) from the box

   [0 1 2 3 4 5 6 7 8 9]

   What about a list of first digits?

4. Someone makes a list showing the first letter of each family name in the phone book, going name by name through the book in order. Is it sensible to model this sequence of letters by drawing at random with replacement from a box? (There would be 26 tickets in the box, each ticket marked with one letter of the alphabet.) Explain.

5. "The smart professional gambler, when heads comes up four times in a row, will bet that it comes up again. A team that's won six in a row will win seven. *He believes in the percentages.* The amateur bettor will figure that heads can't come up again, that tails is 'due.' He'll bet that a team on a losing streak is 'due' to win. *The amateur believes in the law of averages.*"

   —Jimmy the Greek, *San Francisco Chronicle*, July 2, 1975

   Kerrich's coin (chapter 16) will be tossed until it lands heads four times in a row. Suppose Jimmy the Greek offers 5 to 4 that the coin will land heads on the next toss. (On heads, he pays you $5; on tails, you pay him $4.) Do you take the bet?

The answers to these exercises are on p. A89.

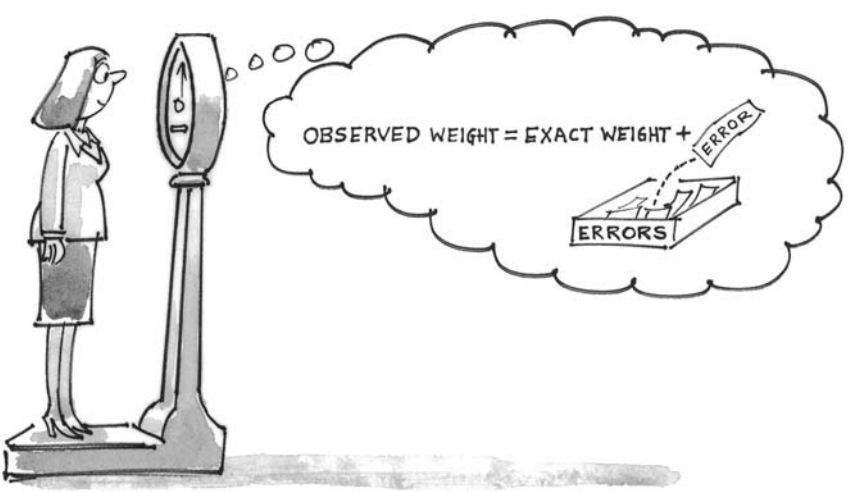
3. THE GAUSS MODEL

The box model for measurement error will now be described in more detail. The basic situation is that a series of repeated measurements are made on some quantity. According to the model, each measurement differs from the exact value by a chance error; this error is like a draw made at random from a box of tickets—the *error box*. Successive measurements are done independently and under the same conditions, so the draws from the error box are made with replacement. To capture the idea that the chance errors aren't systematically positive or systematically negative, it is assumed that the average of the numbers in the error box equals 0. This model is named after Carl Friedrich Gauss (Germany, 1777–1855), who worked on measurement error in astronomical data.

> In the Gauss model, each time a measurement is made, a ticket is drawn at random with replacement from the error box. The number on the ticket is the chance error. It is added to the exact value to give the actual measurement. The average of the error box is equal to 0.

In the model, it is the SD of the box which gives the likely size of the chance errors. Usually, this SD is unknown and must be estimated from the data. Take the 100 measurements on NB 10, for example. According to the model, each measurement is around the exact weight, but it is off by a draw from the error box:

$$\begin{aligned}
\text{1st measurement} &= \text{exact weight} + \text{1st draw from error box}\\
\text{2nd measurement} &= \text{exact weight} + \text{2nd draw from error box}\\
&\;\;\vdots\\
\text{100th measurement} &= \text{exact weight} + \text{100th draw from error box}
\end{aligned}$$



Carl Friedrich Gauss (Germany, 1777–1855)

Wolff-Leavenworth collection, courtesy of the
Syracuse University Art Collection.

With the NB 10 data, the SD of the 100 draws would be a fine estimate for the SD of the error box.[5] The catch is that the draws cannot be recovered from the data, because the exact weight is unknown. However, the variability in the measurements equals the variability in the draws, because the exact weight does not change from measurement to measurement. More technically, adding the exact value to all the errors does not change the SD (pp. 92–93). That is why statisticians use the SD of the measurements when computing the SE. And that completes the reasoning behind the calculation in section 1.[6]

> When the Gauss model applies, the SD of a series of repeated measurements can be used to estimate the SD of the error box. The estimate is good when there are enough measurements.

There may be another way to get at the SD of the error box. When there is a lot of experience with the measurement process, it is better to estimate the SD from all the past data rather than a few current measurements. The reason: the error box belongs to the measurement process, not the thing being measured.

Example 5. (Hypothetical.) After making several hundred measurements on NB 10 and finding the SD to be about 6 micrograms, the Bureau's investigators misplace this checkweight. They go out and buy a new one. They measure its weight by exactly the same procedure as for NB 10, and on the same scale. After a week, they accumulate 25 measurements. These average out to 605 micrograms above 10 grams, and the SD is 7 micrograms. Assuming the Gauss model, the new weight is 605 micrograms above 10 grams, give or take about

6 micrograms 7 micrograms 1.2 micrograms 1.4 micrograms.

Solution. According to the model, the chance error in each measurement is like a draw from the error box. The error box belongs to the scales, not the weight. The SD of the error box should be estimated by the SD of the large amount of past data on NB 10, not the small amount of current data on the new weight. The SD of the error box is estimated as 6 micrograms. This tells the likely size of the chance error in a single measurement. But the likely size of the chance error in the average of 25 measurements is smaller. The SE for the average is 1.2 micrograms. That is the answer.

In the model, the error box belongs to the scales, not the weight. This seems reasonable for chunks of metal which are similar in size. However, if we change from a 10-gram weight to a 100-gram weight, the error box could change too. And for weights that wiggle around more actively—like babies—the separation between "true value" and "chance error" might not be so convincing.

A final point. The version of the Gauss model presented here makes the assumption that there is no bias in the measuring procedure. When bias is present, each measurement is the sum of three terms:

$$\text{exact value} + \text{bias} + \text{chance error}.$$

Then the SE for the average no longer says how far the average of the measurements is from the exact value, but only how far it is from

$$\text{exact value} + \text{bias}.$$

The methods of this chapter are no help in judging bias. We did not take it into account for the measurements on NB 10, because other lines of reasoning suggest that the bias in precision weighing at the National Bureau of Standards is negligible. In other situations, bias can be more serious than chance errors—and harder to detect.[7]

Exercise Set C

1. (a) A 10-gram checkweight is being weighed. Assume the Gauss model with no bias. If the exact weight is 501 micrograms above 10 grams, and the number drawn from the error box is 3 micrograms, what would the measurement be?
   (b) Repeat, if the exact weight is 510 micrograms above 10 grams, and the number drawn from the error box is −6 micrograms.

2. The first measurement on NB 10 was 409 micrograms below 10 grams. According to the Gauss model (with no bias),

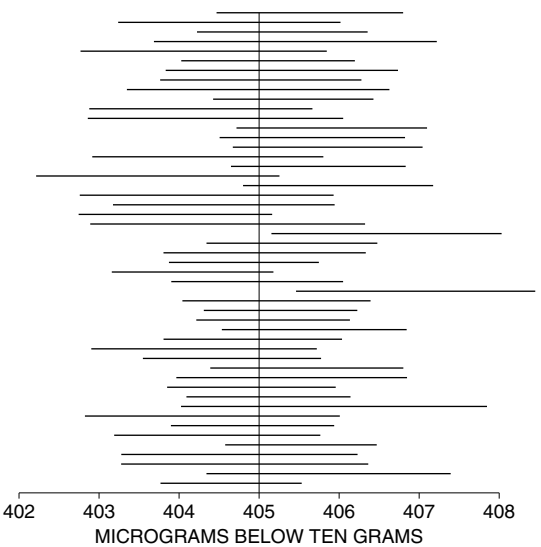
$$409 = \text{exact value} + \text{chance error}.$$

   Can you figure out the numerical value for each of the two terms? Explain briefly.

3. In the Gauss model for the measurements on NB 10, the SD of the error box is _____ 6 micrograms. Fill in the blank using one of the two phrases below, and explain briefly.

   known to be estimated from the data as

4. The figure below shows the result of a computer simulation: 50 imaginary investigators set out to weigh NB 10, following the procedure used by the Bureau. Each investigator takes 100 measurements and computes the average, the SD, and the SE for the average. The 50 confidence intervals "average ± 2 SE" are plotted at different heights in the figure so they can be seen. In the simulation, the exact weight is taken as 405 micrograms below 10 grams.
   (a) Why do the intervals have different centers?
   (b) Why do they have different lengths?
   (c) How many should cover the exact weight?
   (d) How many do?



5. The Bureau is about to weigh a one-kilogram checkweight 100 times, and take the average of the measurements. They are willing to assume the Gauss model, with no bias, and on the basis of past experience they estimate the SD of the error box to be 50 micrograms.
   (a) The average of all 100 measurements is likely to be off the exact weight by _____ or so.
   (b) The SD of all 100 measurements is likely to be around _____.
   (c) Estimate the probability that the average of all 100 measurements will be within 10 micrograms of the exact weight.

6. Suppose you sent a nominal 10-gram weight off to the Bureau, asking them to weigh it 25 times and tell you the average. They will use the same procedure as on NB 10, where the SD of several hundred measurements was about 6 micrograms. The 25 measurements average out to 307 micrograms above 10 grams, and the SD is about 5 micrograms. Your weight is 307 micrograms above 10 grams, give or take around

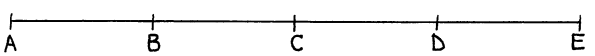
   5 micrograms 6 micrograms 1 microgram 1.2 micrograms

   (You may assume the Gauss model, with no bias.)

7. Twenty-five measurements are made on the speed of light. These average out to 300,007 and the SD is 10, the units being kilometers per second. Fill in the blanks in part (a), then say whether each of (b–f) is true or false. Explain your answers briefly. (You may assume the Gauss model, with no bias.)

   (a) The speed of light is estimated as _____. This estimate is likely to be off by _____ or so.
   (b) The average of all 25 measurements is off 300,007 by 2 or so.
   (c) Each measurement is off 300,007 by 10 or so.
   (d) A 95%-confidence interval for the speed of light is $300{,}007 \pm 4$.
   (e) A 95%-confidence interval for the average of the 25 measurements is $300{,}007 \pm 4$.
   (f) If a 26th measurement were made, there is a 95% chance that it would be off the exact value for the speed of light by less than 4.

8. A surveyor is measuring the distance between five points A, B, C, D, E. They are all on a straight line. She finds that each of the four distances AB, BC, CD, and DE measures one mile, give or take an inch or so. These four measurements are made independently, by the same procedure.



   The distance from A to E is about four miles; but this estimate is likely to be off by around

   4 inches 2 inches 1 inch 1/2 inch 1/4 inch.

   Explain briefly. (You may assume the Gauss model, with no bias.)

9. The concept of measurement error is often applied to the results of psychological tests. The equation is

$$\text{actual test score} = \text{true test score} + \text{chance error}.$$

   The chance error term reflects accidental factors, like the mood of the subject, or luck. Do you think that the Gauss model applies?

The answers to these exercises are on pp. A89–90.

4. CONCLUSION

NB 10 is just a chunk of metal. It is weighed on a contraption of platforms, gears, and levers. The results of these weighings have been subjected to a statistical analysis involving the standard error, the normal curve, and confidence intervals. It is the Gauss model which connects the mathematics to NB 10. The chance errors are like draws from a box; their average is like the average of the draws. The number of draws is so large that the probability histogram for the average will follow the normal curve very closely. Without the model there would be no box, no standard error, and no confidence levels.

Statistical inference uses chance methods to draw conclusions from data. Attaching a standard error to an average is an example. Now it is always possible to go through the SE procedure mechanically. Many computer programs will do the work for you. It is even possible to label the output as a "standard error."

Do not get hypnotized by the arithmetic or the terminology. The procedure only makes sense because of the square root law. The implicit assumption is that the data are like the results of drawing from a box (an old point, but worth repeating). Many investigators don't pay attention to assumptions. The resulting "standard errors" are often meaningless.[8]

> Statistical inference can be justified by putting up an explicit chance model for the data. No box, no inference.

Parts II and III focused on *descriptive statistics*—drawing diagrams or calculating numbers which summarize data and bring out the salient features. Such techniques can be used very generally, because they do not involve any hidden assumptions about where the data came from. For statistical inference, however, models are basic.

5. REVIEW EXERCISES

1. Laser altimeters can measure elevation to within a few inches, without bias, and with no trend or pattern to the measurements. As part of an experiment, 25 readings were made on the elevation of a mountain peak. These averaged out to 81,411 inches, and their SD was 30 inches. Fill in the blanks in part (a), then say whether each of (b–f) is true or false. Explain your answers briefly.
   (a) The elevation of the mountain peak is estimated as _____; this estimate is likely to be off by _____ or so.
   (b) 81,411 ± 12 inches is a 95%-confidence interval for the elevation of the mountain peak.
   (c) 81,411 ± 12 inches is a 95%-confidence interval for the average of the 25 readings.
   (d) There is about a 95% chance that the next reading will be in the range 81,411 ± 12 inches.
   (e) About 95% of the readings were in the range 81,411 ± 12 inches.
   (f) If another 25 readings are made, there is about a 95% chance that their average will be in the range 81,411 ± 12 inches.

2. The first measurement on NB 10 was 409 micrograms below 10 grams. The average of all 100 measurements was 404.6 micrograms below 10 grams, with an SD of 6.4 micrograms; the data are shown in figure 2 on p. 102. Fill in the blanks with a word or phrase. Explain briefly. You may assume the Gauss model, with no bias.
   (a) $404.6 \pm 2 \times 0.64$ is a 95%-confidence interval for the weight of NB 10 because, with 100 draws from the box, the _____ follows the normal curve.
   (b) $409 \pm 2 \times 6.4$ isn't a 95%-confidence interval for the weight of NB 10 because the _____ doesn't follow the normal curve.

3. The speed of light was measured 2,500 times. The average was 299,774 kilometers per second, and the SD was 14 kilometers per second.[9] Assume the Gauss model, with no bias. Find a 95%-confidence interval for the speed of light.

4. In exercise 3, light was timed as it covered a certain distance. The distance was measured 57 times, and the average of these measurements was 1.594265 kilometers. What else do you need to know to decide how accurate this value is?

5. Exercise 4 points to one possible source of bias in the measurements described in exercise 3. What is it?

6. In 2005, the average of the daily maximum temperature at San Francisco airport was 65.8 degrees, and the SD was 7.0 degrees (figure 1, p. 447). Now

$$\sqrt{365} \times 7.0 \approx 134 \text{ degrees}, \qquad 134/365 \approx 0.4 \text{ degrees}.$$

True or false: a 95%-confidence interval for the average daily maximum temperature at San Francisco airport is 65.8 ± 0.8 degrees. Explain briefly.

7. A calibration laboratory has been measuring a one-kilogram checkweight by the same procedure for several years. They have accumulated several hundred measurements, and the SD of these measurements is 18 micrograms. Someone now sends in a one-kilogram weight to be calibrated by the same procedure. The lab makes 50 measurements on the new weight, which average 78.1 micrograms above a kilogram, and their SD is 20 micrograms. If possible, find a 95%-confidence interval for the value of this new weight. (You may assume the Gauss model, with no bias.)

8. In a long series of trials, a computer program is found to take on average 58 seconds of CPU time to execute, and the SD is 2 seconds. There is no trend or pattern in the data. It will take about _____ seconds of CPU time to execute the program 100 times, give or take _____ seconds or so. (The CPU is the "central processing unit," where the machine does logic and arithmetic.)

9. A machine makes sticks of butter whose average weight is 4.0 ounces; the SD of the weights is 0.05 ounces. There is no trend or pattern in the data. There are 4 sticks to a package.
   (a) A package weighs _____, give or take _____ or so.
   (b) A store buys 100 packages. Estimate the chance that they get 100 pounds of butter, to within 2 ounces.

10. True or false, and explain: "If the data don't follow the normal curve, you can't use the curve to get confidence levels."

11. "All measurements were made twice. If two staff members were present, the duplicate measurements were made by different people. In order to minimize gross errors, discrepancies greater than certain arbitrary limits were measured a third time, and if necessary a fourth, until two measurements were obtained which agreed within the set limits. In cases of discrepancy, the mea-

surers decided which of the three or four results was most 'representative' and designated it for inclusion in the statistical record. In cases of satisfactory agreement, the statistical record was based routinely on the first measurement recorded." Comment briefly.[10]

6. SUMMARY

1. According to the *Gauss model* for measurement error, each time a measurement is made, a ticket is drawn at random with replacement from the *error box*. The number on the ticket is the chance error. It is added to the exact value of the thing being measured, to give the actual measurement. The average of the error box is equal to 0. Here, bias is assumed to be negligible.

2. When the Gauss model applies, the SD of many repeated measurements is an estimate for the SD of the error box. This tells the likely size of the chance error in an individual measurement.

3. The average of the series is more precise than any individual measurement, by a factor equal to the square root of the number of measurements. The calculation assumes that the data follow the Gauss model.

4. An approximate confidence interval for the exact value of the thing being measured can be found by going the right number of SEs either way from the average of the measurements; the confidence level is taken from the normal curve. The approximation is good provided the Gauss model applies, with no bias, and there are enough measurements.

5. With the Gauss model, the chance variability is in the measuring process, not the thing being measured. The word "confidence" is to remind you of this.

6. If the model does not apply, neither does the procedure for getting confidence intervals. In particular, if there is any trend or pattern in the data, the formulas may give silly answers.

7. *Statistical inference* is justified in terms of an explicit *chance model* for the data.